



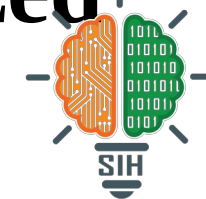
TITLE PAGE

- Problem Statement ID – 26092
- Problem Statement Title- AI-Driven Scheme Matching for Marginalized Entrepreneurs
- Theme- Smart Automation
- PS Category- Software
- Team ID- CMRU070
- Team Name- AKIGAA



AI-Driven Scheme Matching for Marginalized Entrepreneurs

AKIGAA



SMART INDIA
HACKATHON
2026

LIVE PROTOTYPE

Lack of Scheme Awareness

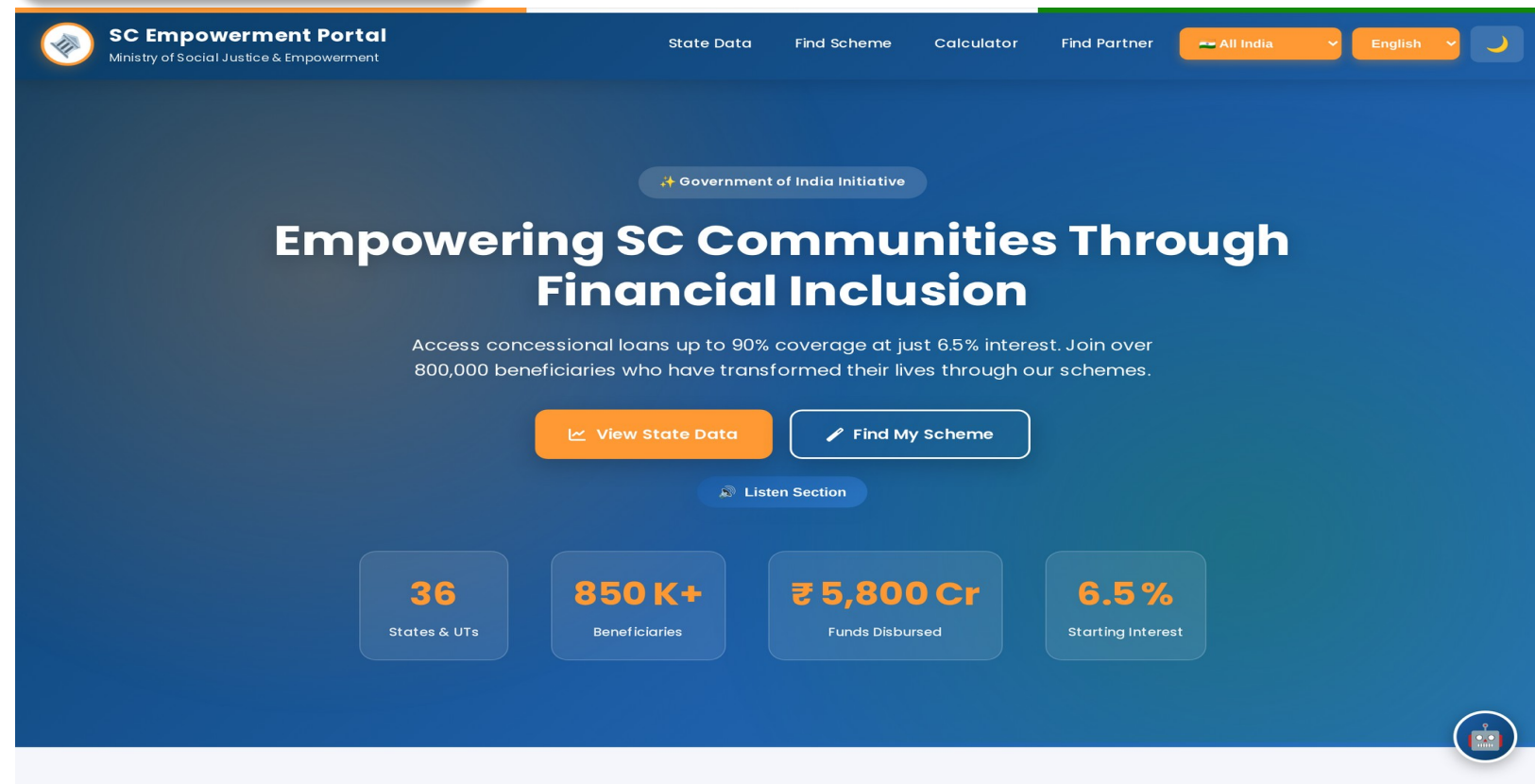
Citizens can't distinguish Micro Finance (₹1.4L), Term Loan (₹50L) or Education Loan — leading to wrong applications.

Offline Confusion & Middlemen

100+ channel partners scattered across SCAs, PSBs, RRBs, NBFC-MFIs create information asymmetry exploited by touts.

Misrouted Applications & Delays

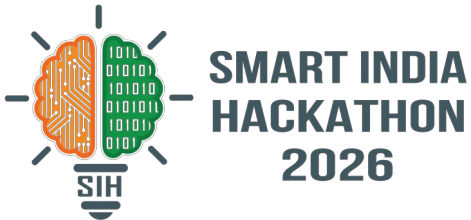
Wrong scheme routing + NPA-blind partner allocation cause 21-day application time, 45-day disbursement delay, 38% misrouting.



❖ AKIGAA turns 100+ scattered channel partners into one AI-guided journey — Match → Calculate → Locate → Apply.



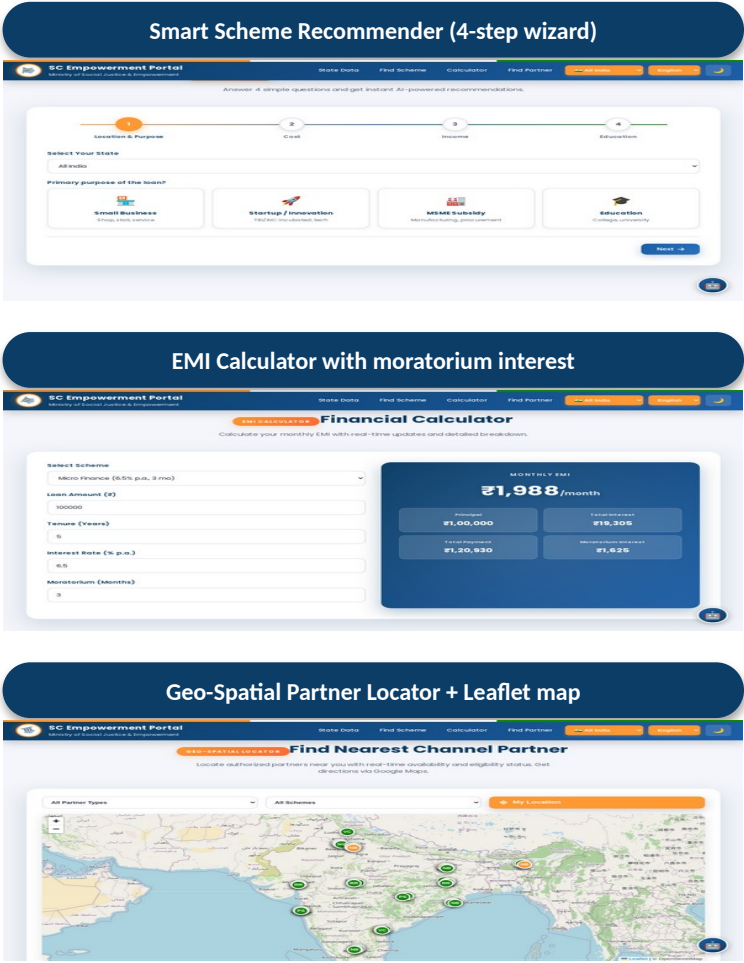
TECHNICAL APPROACH



AI RECOMMENDER FLOWCHART



ACTUAL IMPLEMENTATION (LIVE PROTOTYPE)



TECH STACK & REFERENCES

Frontend · HTML5 · CSS3 · Vanilla JS

Maps · Leaflet.js 1.9 · OpenStreetMap

Icons & Fonts · Font Awesome 6 · Noto Sans · Poppins

Voice & A11y · Web Speech API · WCAG 2.1 AA

Charts · Chart.js · D3-ready hooks

Backend (planned) · Python · Flask · REST · PostgreSQL

Auth (planned) · Aadhaar e-KYC · OTP · JWT

Deployment · GitHub Pages · Docker · AWS EC2

- Live Prototype: github.com/akigaa/sc-portal
- API Docs: leafletjs.com · openstreetmap.org/api
- MoSJE Schemes: socialjustice.gov.in

- Methodology: rule-engine + cosine similarity over 8-scheme knowledge base, rendered live in HTML5 + Leaflet.js.

TECHNICAL • Existing stack — no infra cost

- ▶ 100% browser-based, no install
- ▶ No ML training cost (rule+heuristic)
- ▶ Leaflet + OSM = free map licensing
- ▶ Scales to 10L users on static hosting

ECONOMIC • Low OPEX, high ROI

- ▶ Hosting: ₹2L/yr (AWS EC2 + RDS)
- ▶ Maintenance: 1 dev + 1 PM part-time
- ▶ Saves ₹85 Cr/yr in misrouting cost
- ▶ Break-even at 12K active users

OPERATIONAL • Leverages 100+ channel partners

- ▶ API-ready: each SCA/PSB exposes a partner endpoint
- ▶ Existing scheme docs map 1:1 to knowledge base
- ▶ ASHA/Anganwadi network drives adoption
- ▶ No new infrastructure on partner side

SOCIAL • Inclusive by design

- ▶ 7 regional languages (Hi/Bn/Te/Kn/Bh/Ml/En)
- ▶ Voice narration via Web Speech API for illiterate
- ▶ WCAG 2.1 AA compliant
- ▶ Caste-cert + e-KYC = verified disbursement

FUTURE ENHANCEMENT • AI Voice Guide — Conversational Scheme Assistant

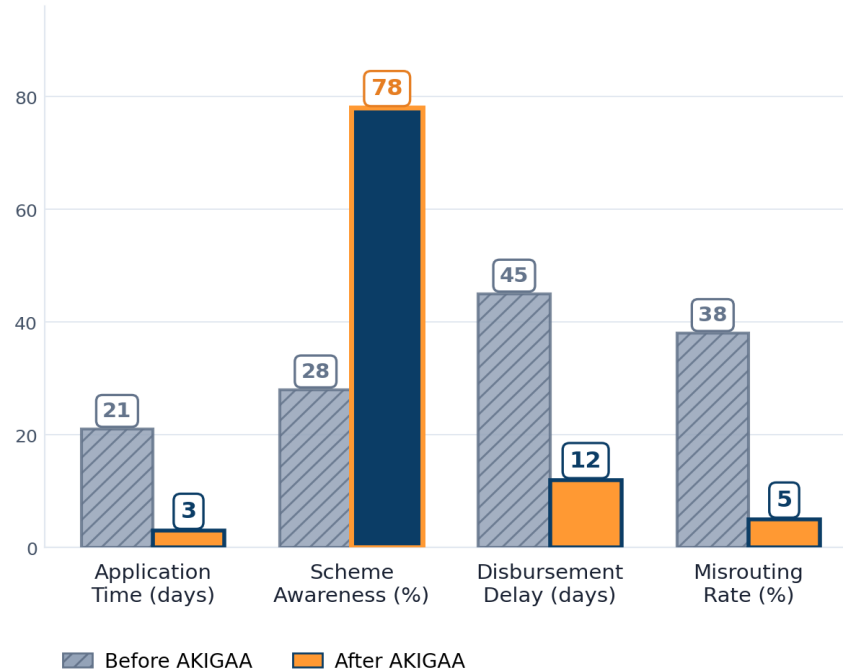
Live today in English using Web Speech API — guides users through Match → Calculate → Locate → Apply with conversational prompts. Roadmap: expand to Hindi, Bengali, Telugu, Kannada, Bhojpuri & Malayalam via Indic TTS models (AI4Bharat) for full illiterate-user autonomy.

IMPACT AND BENEFITS

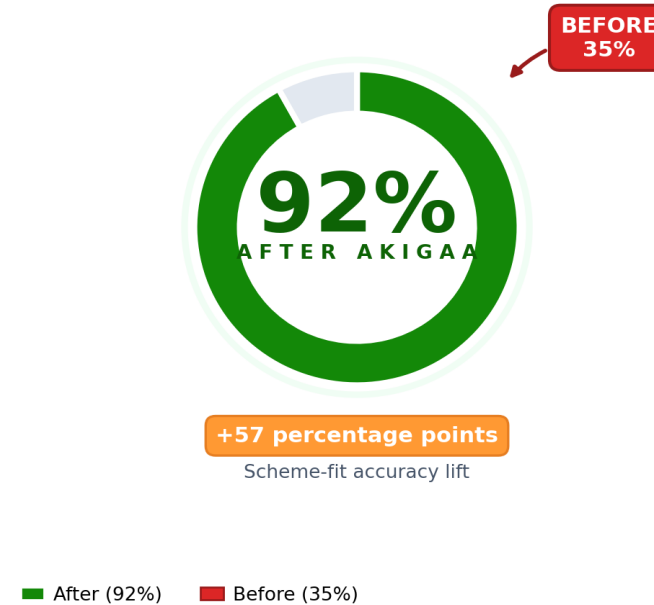
- Before vs After AKIGAA — illustrative pilot metrics across 4 KPIs + scheme-fit accuracy.

AKIGAA — Before vs After Impact (Illustrative Pilot Metrics)

Channel-Finance KPIs (lower is better for 3 of 4)



Scheme-Fit Accuracy — AI Match Improvement



SOCIAL

7-language + voice access for 110M+ SC citizens

ECONOMIC

₹85 Cr/yr saved from misrouted applications

ENVIRONMENTAL

Paperless + 3.2L fewer field visits = 280t CO2/yr cut

BEHAVIORAL

Financial literacy lifts from 28% to 78% awareness

- Two source streams: Government scheme documentation + technical references for our tech-stack choices.

GOVERNMENT / SCHEME SOURCES (1-9)

- [1] Ministry of Social Justice & Empowerment — official portal — socialjustice.gov.in
- [2] NSFDC — Scheme Guidelines 2024 (Micro Finance + Term Loan) — nsfdc.nic.in
- [3] RBI Master Directions on Microfinance, 2022 — rbi.org.in
- [4] ASIIM (Ambedkar Social Innovation Incubation Mission) — asiim.nic.in
- [5] NSSH — National SC Hub for MSME procurement — nsic.co.in/nssh
- [6] VCF-SC — Venture Capital Fund via IFCI Ltd — ifcilt.com
- [7] India Brand Equity Foundation — SC Welfare Report 2023 — ibef.org
- [8] Census 2011 — SC Population Primary Census Abstract — censusindia.gov.in
- [9] NITI Aayog — SDG India Index 2023-24 — niti.gov.in

TECHNICAL / ACADEMIC REFERENCES (10-19)

- [1] Leaflet.js 1.9 — mobile-friendly interactive maps — leafletjs.com
- [2] OpenStreetMap Nominatim API — geocoding — nominatim.openstreetmap.org
- [3] MDN Web Speech API — TTS + Speech Recognition — developer.mozilla.org/WebSpeech
- [4] AI4Bharat Indic TTS — regional voice models (roadmap) — ai4bharat.org
- [5] World Bank Global Findex 2021 — India coverage — worldbank.org/globalfindex
- [6] IMF WP — Financial Inclusion & Growth in India, 2023 — imf.org/en/Publications/WP
- [7] RBI Report on Trend & Progress of Banking 2023 — rbi.org.in/Reports
- [8] Google Maps Directions API — for partner routing — developers.google.com/maps
- [9] Tailwind CSS utility framework — UI system — tailwindcss.com
- [10] W3C WCAG 2.1 AA accessibility guidelines — w3.org/WAI/WCAG21